

Minseok Kim

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EDUCATION

Ajou University

Mar 2021 – Feb 2027

B.S. in Software and Computer Engineering (expected)

GPA: **4.25/4.50** · Major GPA: **4.22/4.50**

RESEARCH INTERESTS

Edge AI System, LLM Inference Optimization, Vision-Language Models, On-Device AI,
GPU Computing, Algorithm

SKILLS

C/C++, Python, Swift, SQL, PyTorch

EXPERIENCE

iKnow Lab., Ajou University

Jul 2025 – Mar 2026

Undergraduate Researcher · Advisor: Prof. Hyunsouk Cho

- Designed controlled experiments and synthetic-data generation pipelines to study spatial reasoning in vision-language models.
- Implemented multi-model inference and attention analysis, and participated in regular lab seminars.

AI·SW Maestro, 17th Cohort (Seoul)

Apr 2026 – Present

Trainee · On-Device AI Development

- Developing Byeolmuri, an on-device AI character app for iOS, in a team-based training program, with responsibility for AI development, on-device evaluation, and improvements informed by TestFlight user feedback.

Republic of Korea Army

Mar 2022 – Sep 2023

CERT Soldier · Cyber Network Operations Center (CNOC)

- Monitored network access control, blocked unauthorized devices, and handled security events at the Armed Forces Command and Communications Command.
- Completed mandatory service as a Sergeant.

PUBLICATION

Diagnosing Axis-Dependent Spatial Bias in Vision-Language Models

Minseok Kim, Hyunsouk Cho

Korea Computer Congress (KCC 2026), Jeju, South Korea

Undergraduate Paper Competition, Poster Presentation · Jun 2026

- Observed an axis-dependent accuracy gap of up to **52.12 percentage points** across four vision-language models.
- Using **2,000 controlled samples**, found a negative correlation between patch sequence distance and inter-object attention (**Spearman's $\rho = -0.782$**); accuracy partially rebounded at the longest distance.

ACTIVITIES AND PROJECTS

Byeolmuri — On-Device AI Character App

May 2026 – Present

AI Developer · AI · SW Maestro Team Project

- Built an iOS on-device agent with Swift and LiteRT-LM, SQLite-backed memory, and validated tool execution.
- Reduced false tool activation from **45.83% to 4.17%** through intent classification and embedding-based tool selection, maintaining approximately **94%** intent classification performance on the evaluation set.
- Reduced mean time to first token by **74.1% (4.920 s to 1.274 s)** through fixed-prompt KV cache reuse on iPhone 17 CPU across 30 requests, with matching outputs in all 15 response pairs.
- Integrated device memory measurements into the evaluation harness and added streaming responses following TestFlight feedback.

TOWA — AI Manga Translation Workstation

Mar 2026 – Jun 2026

AI Developer · University Team Project

- Built the Model Engine connecting text detection, OCR, LLM translation, and inpainting; preserved intermediate outputs to trace image-quality failures.
- Reduced MSE-based pixel damage from **23.15% to 1.50%** with masked compositing and OpenCV post-processing. Cut end-to-end workflow time by approximately **60%** across three manga comparisons.

A.N.S.I., Ajou University

Mar 2024 – Sep 2026

Member · Algorithm and Programming Club

- Practiced C++ problem solving, joined coding camps and ICPC team training, and competed in SPCP, UCPC, and the ICPC Seoul preliminary contest.
- Placed **21st** in the 2025 shake! finals as an Ajou University representative (held Jan 2026).

HONORS AND SCHOLARSHIPS

Third Prize (5th overall)

Sep 2026

5th Inter-University Deep Learning Challenge

Ajou University

Honorable Mention

May 2026

25th TOPCIT, Ajou University / AI Convergence Education Institute Director's Award

Excellence Award

May 2026

Ajou Programming Contest (APC), Div. 1

Director's Award, Ajou University SW-Centered University Program

Second Prize (3rd overall)

Oct 2025

4th Inter-University Deep Learning Challenge

Ajou University (host); Kyungpook National, Kyung Hee, Pusan National, Chonnam National, and Jeonbuk National Universities (co-organizers)

6th Overall / Ajou University First-Place Special Award

Sep 2025

Centroid Cup / Kookmin, Chung-Ang, Inha, and Ajou Universities

Excellence Award

May 2025

Ajou Hackathon

Director's Award, Ajou University SW Convergence Education Institute

Two Excellence Awards and One Encouragement Award

2024–2026

Mogakso, Ajou University

Excellence: Winter 2024, Summer 2025 / Encouragement: Summer 2026

Yulgok Academic Merit Scholarship

2021–2026

Ajou University / Four semesters / Approximately US\$6,900 in total
Spring 2021, Spring 2025, Fall 2025, Spring 2026